Materials and methods

Plant material and genotyping

A wild barley (*Hordeum spontaneum*) germplasm collection comprising 300 accessions was established in 2017 and used across the three growing seasons of the present study. The collection was sampled from 30 sites representing diverse environmental and climatic conditions across Israel, capturing gradients in soil type, elevation, precipitation, and temperature. Sampling sites were selected on the basis of previous population-genetic studies of wild barley in Israel, designed to maximize the representation of environmental and genetic diversity while minimizing the confounding effects of demographic history and population structure (Hübner et al., 2009, 2012, 2013; Hübner & Kantar, 2021). The sites encompass the three previously described ecotypes — northern, coastal, and desert — together with the three intermediate ecotones. Within each site, the specific sampling location was chosen to capture local environmental and topographic variation (e.g., differing slopes and microclimates) while avoiding anthropogenic disturbance. Ten individuals were collected per site, with a minimum inter-plant distance of ten meters. All accessions underwent two rounds of single-seed descent (SSD) and selfing. Geographical coordinates were recorded at each sampling site.

All 300 accessions were whole-genome sequenced at an average coverage of 5× on an Illumina NovaSeq 6000 platform, as described in Shermeister (2021). Raw reads were assessed for base quality and GC content using FastQC v0.11.5 (Andrews, 2017), and adapter sequences together with low-quality bases were trimmed using fastp v0.20.0 (Chen et al., 2018). Cleaned reads were aligned to the barley *Hordeum vulgare* cv. Morex reference genome assembly v3 (Mascher et al., 2021) using BWA-MEM2 v2.2.1 (Vasimuddin et al., 2019) under default parameters, and duplicate reads were subsequently marked using Picard MarkDuplicates v3.4.0 (Broad Institute, 2019). Variant discovery was performed following the GATK4 best-practices workflow (Poplin et al., 2017; Heldenbrand et al., 2019) implemented in GATK v4.6.2.0. Per-sample variants were called in GVCF mode using HaplotypeCaller with ploidy set to 2; the resulting GVCFs were condensed with ReblockGVCF, consolidated across accessions using GenomicsDBImport, and jointly genotyped using GenotypeGVCFs with the maximum number of alternate alleles per site limited to two.

Post-genotyping variant filtering was conducted using bcftools v1.13 (Danecek et al., 2021) and VCFtools v0.1.15 (Danecek et al., 2011). Only biallelic SNPs were retained and INDELs were removed. Variants were excluded on the basis of the following hard-filter thresholds: Quality by Depth (QD) < 5, Mapping Quality (MQ) < 45, Fisher Strand bias (FS) > 60, Strand Odds Ratio (SOR) > 3, Mapping Quality Rank Sum Test (MQRankSum) < −2.5, Quality (QUAL) < 140, or total site depth (DP) outside the range 900–1,800. Given the predominantly selfing mating system of H. spontaneum, residual heterozygous calls were set to missing; in addition, individual genotype calls supported by fewer than three reads (FORMAT/DP < 3) were set to missing. The dataset was then filtered for a minimum minor allele frequency (MAF) of 0.05 and a maximum per-site missing data rate of 30%. Following these filters, ten accessions

Commented [של1]: Hübner, S., Höffken, M., Oren, E., Haseneyer, G., Stein, N., Graner, A., Schmid, K., & Fridman, E. (2009). Strong correlation of wild barley (*Hordeum spontaneum*) population structure with temperature and precipitation variation. *Molecular Ecology*, 18(7), 1523–1536. https://doi.org/10.1111/j.1365-294X.2009.04106.x **[From Noam Pintel — carried over; also in evgeny.full.pdf]**
Hübner, S., Bdolach, E., Ein-Gedy, S., Schmid, K. J., Korol, A., & Fridman, E. (2013). Phenotypic landscapes: genetic and environmental drivers of shape variation. *Heredity*, 110(1), 103–112. **[From Noam Pintel — carried over]** ⚠️ See flag below.
Hübner, S., Rashkovetsky, E., Kim, Y. B., Oh, J. H., Michalak, K., Weiner, D., Korol, A. B., Nevo, E., & Michalak, P. (2013). Genome differentiation of *Drosophila melanogaster* from a microclimate contrast in Evolution Canyon, Israel. *PNAS*, 110(52), 21059–21064. ⚠️ **FLAG**

Commented [של2R1]: Hübner et al., 2013 citation — ambiguous. The string "Hübner et al. 2012, 2013" appears in both Noam's thesis and the current draft. I could not resolve with certainty *which* 2013 Hübner paper is intended —

Commented [של3]: Shermeister, B. (2021). *Genomic and Phenotypic Characterization of Wild Barley (Hordeum spontaneum) from Israel.* M.Sc. Thesis, Tel-Hai Academic

Commented [של4R3]: Shermeister (2021) citation format. This is a thesis, not a peer-reviewed paper. Noam's thesis cites it the same way. For a TAG-level submission, reviewers may push back on a thesis citation for *the*

Commented [של5]: Andrews, S. (2017). *FastQC: A quality control tool for high-throughput sequence data.* Babraham Bioinformatics. https://www.bioinformatics.babraham.ac.uk/projects/fastqc

Commented [של6]: Chen, S., Zhou, Y., Chen, Y., & Gu, J. (2018). fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics*, 34(17), i884–i890.

Commented [של7]: Mascher, M., Wicker, T., Jenkins, J., Plott, C., Lux, T., Koh, C. S., et al. (2021). Long-read sequence assembly: a technical evaluation in barley. *The Plant Cell*, 33(6), 1888–1906.

Commented [של8]: Vasimuddin, M., Misra, S., Li, H., & Aluru, S. (2019). Efficient architecture-aware acceleration of BWA-MEM for multicore systems. *2019 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*,

Commented [של9]: Broad Institute. (2019). *Picard Toolkit.* http://broadinstitute.github.io/picard/ **[From Noam Pintel — carried over]**

Commented [של10]: Poplin, R., Ruano-Rubio, V., DePristo, M. A., Fennell, T. J., Carneiro, M. O., Van der Auwera, G. A., et al. (2017). Scaling accurate genetic variant discovery to tens of thousands of samples. *bioRxiv*, 201178

Commented [של11]: *Danecek, P., Auton, A., Abecasis, G., Albers, C. A., Banks, E., DePristo, M. A., Handsaker, R. E., Lunter, G., Marth, G. T., Sherry, S. T., McVean, G., & Durbin, R. (2011). The variant call format and VCFtools.*

Commented [של12]: Danecek, P., Bonfield, J. K., Liddle, J., Marshall, J., Ohan, V., Pollard, M. O., Whitwham, A., Keane, T., McCarthy, S. A., Davies, R. M., & Li, H. (2021). Twelve years of SAMtools and BCFtools. *GigaScience*,

from the Hachola site (Site 04) were removed from the dataset due to poor sample quality, yielding a final genotypic dataset of 290 accessions and 7,110,996 high-quality biallelic SNPs that served as the input for all subsequent genomic analyses.

Field experiment and phenotyping

Common-Garden Field Design and Experimental Layout

Three field experiments were conducted under common-garden conditions across three consecutive growing seasons (2019–2020, 2020–2021, and 2021–2022) in a net-house at the Mataim Farm, Hula Valley, Israel (33°09'08.4"N, 35°37'15.8"E). Prior to sowing, seeds were surface-sterilized in 3% sodium hypochlorite for five minutes, rinsed for five minutes with distilled water, and placed on moist germination paper at 16 °C for 18 days to break dormancy and synchronize germination across accessions. Germinated seedlings were subsequently transplanted into 5-liter pots filled with a commercial planting substrate ("Ram 158", Tuff Merom Golan Ltd., Merom Golan, Israel). Within each season, the 300 accessions were arranged in a randomized complete block design (RCBD), with each accession represented once per block and a total of 300 plants per block. The first two seasons comprised four blocks each (1,200 plants per season), whereas the third season was expanded to five blocks (1,500 plants). Following systematic quality control of the 2019–2020 dataset, Block 4 of that season was excluded from all downstream nutritional-trait analyses due to substantially lower inter-block correlations in grain composition relative to Blocks 1–3 (Supplementary Table S2).

Agromorphological Phenotyping and Flowering-Time Staging

To place grain nutritional variation within the broader context of plant development and to provide morphological reference traits for the heritability analyses, two pre-harvest agromorphological traits — flowering time and number of tillers — were recorded on each plant using the FieldBook Android application v5.0.7 (Rife & Poland, 2014). Flowering time was scored according to the Zadoks cereal growth scale (Zadoks et al., 1974), and was defined operationally as the number of days from transplanting to first spike emergence from the flag-leaf sheath. Number of spike-bearing tillers was recorded per plant at the end of the growing season.

GDD Conversion and Climatic Context

To account for inter-annual climatic variation between the three growing seasons, flowering time in days was converted to Growing Degree Days (GDD), an accumulated thermal-time measure that standardizes developmental progress across environments (McMaster & Wilhelm, 1997). Daily temperature records for the net-house site were obtained from the adjacent Kfar Blum meteorological station (station no. 8473; Schwarts, 2020), and daily GDD was calculated as:

$$Daily\ GDD = \frac{t_{min} + t_{max}}{2} - t_{base}$$

where *t*min and *t*max are the daily minimum and maximum air temperatures (°C) and *t*base is the lower developmental threshold for barley, set to 0 °C. Daily *t*min values below *t*base were set to *t*base prior to summation, following the methodology of McMaster and Wilhelm (1997).

Commented [של13]: **Hachola "poor sample quality" — is this specific enough?** A reviewer may ask *what* was poor: DNA extraction failure? Low coverage? Genotyping error rate? High missing data?

Commented [של14]: Supplementary Material to Add

Supplementary Table S2. Pairwise Pearson correlation coefficients between blocks within the 2019–2020 season for the four grain nutritional traits (protein, starch, beta-glucan, dietary fiber). Values compare each pair of blocks; Block 4 shows systematically lower correlations with Blocks 1–3 (e.g., for beta-glucan: r = 0.328–0.391) relative to the pairwise correlations among Blocks 1–3 (e.g., for beta-glucan: r = 0.670–0.682), justifying exclusion of Block 4 from downstream nutritional-trait analyses.

...

Commented [של15]: Rife, T. W., & Poland, J. A. (2014). Field Book: An open-source application for field data collection on Android. *Crop Science*, 54(4), 1624–1627. https://doi.org/10.2135/cropsci2013.08.0579 **[From Noam Pintel — format corrected]**

Commented [של16]: Zadoks, J. C., Chang, T. T., & Konzak, C. F. (1974). A decimal code for the growth stages of cereals. *Weed Research*, 14(6), 415–421. https://doi.org/10.1111/j.1365-3180.1974.tb01084.x **[From Noam Pintel — carried over]**

Commented [של17]: McMaster, G. S., & Wilhelm, W. W. (1997). Growing degree-days: one equation, two interpretations. *Agricultural and Forest Meteorology*, 87(4), 291–300. **[From My Draft — role narrowed to equation methodology only]**

Commented [של18]: Schwarts, 2020 — Israeli Meteorological Service. **[From Noam Pintel — carried over]** ⚠ Format verification still pending (flagged in Paragraph 1).

Commented [של19]: **The only open question is: what citation do we anchor it to?**
Here's where we stand on the citation options:
Option A — McMaster & Wilhelm (1997): Already cited in the same sentence for the clamping method (setting t_min to t_base when it falls below). This paper discusses the 0 °C convention for cereals broadly. It's peer-reviewed, well-

...

Commented [של20]: McMaster, G. S., & Wilhelm, W. W. (1997). Growing degree-days: one equation, two interpretations. *Agricultural and Forest Meteorology*, 87(4), 291–300. DOI: 10.1016/S0168-1923(97)00027-0

Post-Harvest Processing and Grain Recovery
At the end of the growing season, five representative spikes per plant were selected, individually labeled, bagged, and oven-dried at 35 °C for 30 days to achieve uniform moisture equilibration across all samples prior to post-harvest measurements. Spike length was recorded on the dried spikes. The dried spikes were then threshed — manually in the 2019–2020 season and mechanically using a Haldrup LT-21 thresher (Haldrup, Denmark) in the 2020–2021 and 2021–2022 seasons — after which seeds were separated from straw and chaff using a South Dakota seed blower (Seedburo Equipment Company Ltd.), yielding uniformly cleaned grain samples. For each plant, the cleaned grain from the five harvested spikes was weighed (g). Grain weight and spike length were among the eight traits subsequently included in the heritability and mixed-model framework; grain samples were stored for downstream nutritional composition analysis.

Determination of seed nutrients:

To efficiently quantify grain biochemical composition across the full three-season dataset (approximately 3,900 samples), a near-infrared (NIR) analyzer (DA 7250, Perten Instruments Ltd., courtesy of Equinom Ltd.) was used for non-destructive estimation of protein, starch, beta-glucan, and dietary fiber content. To calibrate the device for wild barley, one accession from each of the 30 sampling sites was randomly chosen and its grain scanned by the NIR instrument, and the resulting spectral profiles were compared against an existing cultivated barley calibration package provided by the manufacturer (Perten Ltd.) as a baseline reference.

The 30 randomly chosen accessions were first scanned by the NIR instrument and their spectral data exported for downstream calibration. The same 30 accessions were then analyzed in the laboratory using commercial enzymatic assay kits to provide reference biochemical values against which the NIR predictions could be calibrated. Beta-glucan and total starch concentrations were determined using the Megazyme Mixed-Linkage Beta-Glucan Assay Kit and the Megazyme Total Starch Assay Kit (K-TSTA) (Megazyme, Ireland), respectively, both adapted for small sample volumes. Prior to analysis, grain samples were milled using an IKA A11 analytical mill and sieved through a 50-mesh screen to obtain fine flour. For each accession, three independent flour subsamples of approximately 200 mg were analyzed and the results averaged. Analyte concentrations were expressed as a percent of solid samples and calculated using the Megazyme Mega-Calc™ online data calculator.

Total protein content was determined using the Kjeldahl digestion–distillation–titration method at the Analytical Biochemistry Laboratory, Tel Hai Academic College, using 200 mg of ground seed per sample. Digestion was performed with sulfuric acid and catalyst, followed by alkalization with NaOH, steam distillation of the liberated ammonia into boric acid, and back-titration with standardized HCl. Total protein content was calculated from the measured nitrogen and expressed as a percentage of sample weight, following the protocol of Beljkaš et al. (2010).

A wild-barley-adapted NIR calibration package was assembled using the manufacturer's cultivated barley package (Perten Ltd.) as the baseline. For protein and starch, the cultivated calibration was retained as a baseline and adjusted using

Commented [של21]: ⚠ **Spike length measurement rule — completely unspecified.** The paragraph says "spike length was recorded on the dried spikes" without defining: (a) from what point to what point, (b) whether awns are included, (c) whether the value is a single measurement or an average of the five spikes. **Please specify or ask Sariel/Noam for the protocol.**

Commented [של22]: can optionally be anchored with a section number (e.g., "see Section 2.X") once the paper's numbering is finalized.

Commented [של23]: ⚠ **Catalog code consistency.** K-TSTA is included (per Noam). K-BGLU is not included (not in Noam). You may want these to match each other — either include both catalog codes or neither. Your call on a future polish pass.

Commented [של24]: **Beljkaš, B., Matić, J., Milovanović, I., Jovanov, P., Mišan, A., & Šarić, L. (2010).** Rapid method for determination of protein content in cereals and oilseeds: validation, measurement uncertainty and comparison with the Kjeldahl method. *Accreditation and Quality Assurance*, 15(10), 555–561. https://doi.org/10.1007/s00769-010-0677-6 **[From Noam Pintel — carried over]**

Commented [של25R24]: ⚠ **Beljkaš et al. 2010 scope.** The Beljkaš paper is primarily about the Dumas combustion method and its comparison with Kjeldahl, not exclusively a Kjeldahl protocol paper. It does validate Kjeldahl, so the citation is defensible. Noam used this exact citation. If Dr. Hübner asks for a stronger Kjeldahl-specific citation, the AACC Method 46-11.02 (Crude Protein — Improved Kjeldahl Method) is a common alternative. For now, retain Beljkaš to match Noam.

Commented [של26R24]: ⚠ **Supplementary option — full Kjeldahl protocol.** If you prefer to preserve the full step-by-step chemistry somewhere (e.g., for the Tel Hai lab's internal documentation or for reviewer supplementary requests), the extended version from previous draft could be placed in Supplementary Methods. Your call — not necessary, but an option.

Supplementary Material (optional)
If you want to move the full chemistry description to Supplementary Methods, I can prepare a "Supplementary Methods S1. Detailed Kjeldahl protocol" block. Let me know.

laboratory reference measurements obtained from the 30 randomly chosen accessions, shifting the predictions onto the wild-barley range. Beta-glucan calibration was developed independently from the Megazyme assay measurements and was incorporated manually into the final calibration package. Fiber content was retained from the cultivated package without species-level adjustment, as no independent laboratory reference data were available for this trait. This single wild-barley-adapted calibration was subsequently applied uniformly to all samples across the three growing seasons. To validate the calibration, NIR-predicted values for the 30 reference accessions were compared with their corresponding laboratory measurements using Pearson correlation analysis, and cross-season consistency of NIR predictions was confirmed by computing pairwise Pearson correlations between trait values across seasons (Supplementary Table S3).

Commented [של27]: (for fiber) — This is Noam's stated reason, and your draft preserves it. I kept it. If Sariel knows a different reason (e.g., lab equipment didn't measure fiber, or it was a budget decision), note it for the final pass.

Commented [של28]:

Commented [של29R28]: Supplementary Table S3 (NIR calibration validation) — legend from new Claude is clean and accurate; you can use it:
NIR calibration validation. Pearson correlation coefficients (r) between NIR-predicted values and laboratory reference measurements across the 30 randomly chosen wild barley accessions used for calibration development, reported for (A) starch, (B) protein, (C) beta-glucan, and (D) fiber.

A	Starch	
	Calibration scans	Season 2020 scans
Kit results	0.785***	0.466**
Season 2020 scans	0.548**	

B	Protein	
	Calibration scans	Season 2020 scan
Kit results	0.915***	0.556**
Season 2020 scans	0.649***	

C	Beta-glucan	
	Calibration scans	Season 2020 scans
Kit results	1.000***	0.514**
Season 2020 scans	0.514**	

D	Fiber	
	Calibration scans	Season 2020 scans
Kit results	-	-
Season 2020 scans	0.526**	

Genetic analyses and genome wide association

Phenotypic Data Preparation for Genetic Analysis

All phenotypic data preparation, mixed-model fitting, and variance-component estimation were performed in R v4.4.3 (R Core Team, 2025). Following the exclusion of Block 4 from the 2019–2020 season described above, the three-season phenotypic dataset was further filtered to ensure concordance with the genotypic dataset: the two cultivated control lines (Morex and Clipper) and all records corresponding to the ten Hachola (Site 04) accessions were removed, matching the 290-accession wild barley panel used for genomic analyses. Each of the eight traits included in the genetic analyses — protein content, starch content, beta-glucan content, dietary fiber content, flowering time, number of tillers, grain weight, and spike length — was then mean-centered on a per-season basis by subtracting the seasonal population mean from each individual observation within that season. The four nutritional traits (protein, starch, beta-glucan, and fiber) constituted the primary targets of the genetic analysis; the remaining four agromorphological traits were included to contextualize the estimated heritabilities across a range of trait biology. Within each model fitting step, trait-specific quality control was applied locally: observations with missing values were excluded for the trait under analysis, and observations deviating by more than three standard deviations from the trait mean were removed as outliers prior to model fitting.

Commented [של30]: R Core Team (2025). *R: A language and environment for statistical computing.* R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/ **[AI Sourced — standard R citation]**

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Broad-Sense Heritability (H²) and BLUP Estimation

For each of the eight traits, broad-sense heritability (H²) and Best Linear Unbiased Predictions (BLUPs) were estimated using a linear mixed model (LMM) fitted with the lme4 R package v1.1.37 (Bates et al., 2015). For each trait independently, genotype and season-by-block combination were fitted as crossed random effects, where the season-by-block term indexes each unique combination of growing season and block. Variance components were extracted from the fitted model using the insight R package v1.3.0 (Lüdecke et al., 2019), and broad-sense heritability was calculated as:

$$H^2 = \frac{V_G}{V_G + V_E}$$

where V_G is the variance attributable to genotype and V_E is the sum of the remaining variance components (season-by-block and residual). Genotype-level predicted values were extracted from the same fitted model as the sum of the overall model intercept and the genotype-specific random-effect deviation; these values served as the phenotypic input for all subsequent genomic analyses. Prior to each model fit, the trait-specific local quality control described above (missing-value exclusion and ±3 SD outlier removal) was applied.

Genotype × Environment Interaction Analysis

To characterize the extent to which genotypic responses for grain nutritional traits were consistent across growing seasons, a second linear mixed model was fitted for each trait, extending the model described above by adding a genotype-by-season interaction term as an additional random effect. Variance components were extracted via lme4::VarCorr() and partitioned into four components: genetic variance (V_G), season-by-block environmental variance (V_E), genotype-by-season interaction variance (V_{GxE}), and residual variance (V_R). Heritability was recalculated as:

$$H^2 = \frac{V_G}{V_G + V_E + V_{GxE} + V_R}$$

This model was used exclusively to generate variance partitioning plots and reaction-norm visualizations across the three growing seasons, while the H² values reported for the eight traits (Section X.X) were derived from the model without the interaction term described in the previous paragraph.

Genome-Wide Association Study (GWAS)

To identify genomic regions associated with grain nutritional traits, a genome-wide association study (GWAS) was conducted for each of the four nutritional traits — protein, starch, beta-glucan, and dietary fiber — using the EMMAX software (Efficient Mixed-Model Association eXpedited; Kang et al., 2010). All genomic data handling and downstream statistical analyses were performed in R v4.1.2 (R Core Team, 2021). The BLUP values estimated in the previous section served as the phenotypic input, and the analysis was performed on the full filtered SNP dataset described above, without further variant-level filtering at this step. Population structure was corrected by including the first five principal components (PCs) as fixed covariates, derived using PLINK v1.9 v1.90b6.4 (Purcell et al., 2007) from a linkage-disequilibrium-pruned SNP subset obtained with --indep-pairwise 50 5 0.2. A

Commented [של32]: **Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015).** Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48. https://doi.org/10.18637/jss.v067.i01 **[From My Draft — standard lme4 citation]**

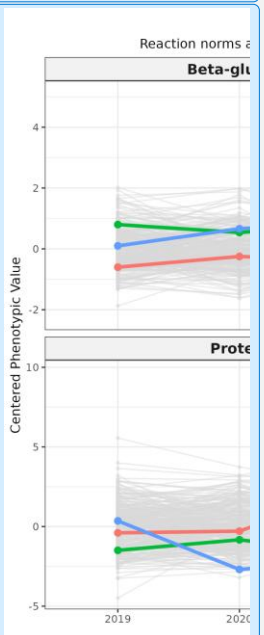
Commented [של33]: **Lüdecke, D., Waggoner, P., & Makowski, D. (2019).** insight: A unified interface to access information from model objects in R. *Journal of Open Source Software*, 4(38), 1412. https://doi.org/10.21105/joss.01412

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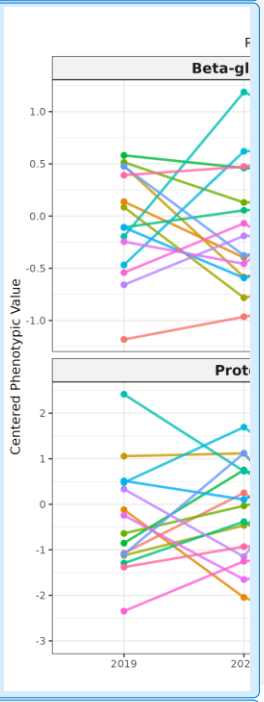
Commented [של35]: ⚠️ **"Section X.X" placeholder.**

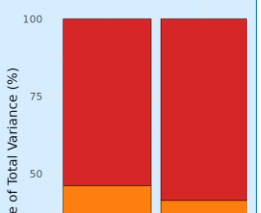
Commented [של36]: Decide later which reaction plots to show.

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pairwise Identity-By-State (IBS) kinship matrix was computed from the same LD-pruned subset using emmax-kin and incorporated as a random effect to model residual genetic relatedness.

To identify significant marker–trait associations, three significance thresholds were computed and visualized for each trait: a genome-wide Bonferroni-corrected threshold ($\alpha = 0.05/n$, where n is the total number of tested SNPs); a Benjamini–Hochberg false discovery rate (FDR) threshold at the 5% level (Benjamini & Hochberg, 1995), drawn only when at least one SNP attained BH significance; and a fixed suggestive threshold of $-\log_{10}(P) > 6$. Genomic regions were considered associated with a trait when at least one SNP exceeded the $-\log_{10}(P) > 6$ threshold and the surrounding signal showed a consistent peak shape in the Manhattan plot, rather than an isolated single-SNP signal; lead SNPs within each region were selected manually based on these criteria. Manhattan plots were generated using the qqman R package v0.1.9 (Turner, 2018), and quantile–quantile (QQ) plots were generated in R, with the genomic inflation factor (λ) computed for each trait to assess model calibration and residual population stratification.

Linkage Disequilibrium Decay and Candidate Gene Window

To determine an empirically justified genomic window for candidate gene identification around each GWAS lead SNP, genome-wide linkage disequilibrium (LD) decay was characterized across the 290-accession collection. Pairwise r^2 values were calculated across all seven barley autosomes (chromosomes 1H–7H) using PLINK v1.90b6.4 (Purcell et al., 2007) on the full filtered SNP dataset, with a maximum physical window of 5,000 kb, a maximum of 999,999 intervening SNPs per comparison, and a minimum r^2 reporting threshold of 0.01. Pairwise r^2 values were aggregated into 1-kb physical distance bins, and the mean r^2 per bin was computed. The decay profile was then visualized in R by fitting a locally estimated scatterplot smoothing (LOESS) curve to the binned means. The critical LD decay distance was defined as the physical distance at which the LOESS-fitted curve intersected an r^2 threshold of 0.2. Based on this analysis, LD decayed to $r^2 = 0.2$ at approximately 220 kb, establishing a ±220 kb search window around each lead SNP for subsequent candidate gene discovery (Figure X.X).

Candidate Gene Identification and VCF Extraction

To identify candidate genes within each associated genomic region, a window-based search was performed using an in-house shell script built around bedtools intersect v2.30.0 (Quinlan & Hall, 2010). For each manually curated lead SNP, a genomic interval of ±220 kb was constructed and intersected with the Morex V3 reference genome annotation (GFF3 format; Ensembl Plants release 62; Yates et al., 2022), restricted to annotated gene features, yielding a list of all genes whose coordinates overlapped the defined window around each lead SNP. For each identified candidate gene, a gene-level VCF was subsequently extracted from the full filtered genotypic dataset using bcftools view v1.13 (Danecek et al., 2021) with the gene's annotated start and end coordinates extended by 1 kb upstream and downstream, thereby capturing proximal regulatory regions alongside the coding sequence. This resulted in one sub-VCF per candidate gene, retaining the full set of segregating variants within each gene's local genomic context, which served as the input for the downstream haplotype analysis.

Commented [של43]: Turner, S. D. (2018). qqman: an R package for visualizing GWAS results using Q-Q and Manhattan plots. *Journal of Open Source Software*, 3(25), 731. https://doi.org/10.21105/joss.00731 **[From My Draft]**

Commented [של44]: Three significance thresholds declared, only one used for lead-SNP calling. The GWAS paragraph computes three thresholds (Bonferroni, FDR at X%, suggestive $-\log_{10} P > 6$), then says "Genomic regions were considered associated with a trait when at least one SNP exceeded the $-\log_{10}(P) > 6$ threshold." The Bonferroni and FDR thresholds are computed/visualized but are not gating criteria.

Commented [של45R44]: Optional change:

For each trait, three significance thresholds were computed and overlaid on the Manhattan plots for reference: a genome-wide Bonferroni-corrected threshold ($\alpha = 0.05/n$, where n is the total number of tested SNPs); a False Discovery Rate (FDR) threshold at the **X%** level, computed using the Benjamini–Hochberg procedure (Benjamini & Hochberg, 1995); and a fixed suggestive threshold of $-\log_{10}(P) > 6$. The Bonferroni and FDR lines are shown purely as visual reference and were not used as gating criteria, as the restricted number of tested individuals (n = 290) renders these thresholds overly conservative for discovery in a collection of this size (Zhang et al., 2019). Instead, genomic regions were operationally defined as trait-associated when at least one SNP exceeded the fixed suggestive threshold of $-\log_{10}(P) > 6$ and the surrounding signal formed a consistent peak-shaped cluster in the Manhattan plot, rather than an isolated single-SNP spike; lead SNPs within each such region were then selected manually on the basis of these criteria.

Commented [של46R44]: Choose 0.05 or 0.1

Commented [של47]: Purcell, S., Neale, B., Todd-Brown, K., Thomas, L., Ferreira, M. A. R., Bender, D., Maller, J., Sklar, P., de Bakker, P. I. W., Daly, M. J., & Sham, P. C. (2007). PLINK: A tool set for whole-genome association and population-based linkage analyses. *American Journal of Human Genetics*, 81(3), 559–575. https://doi.org/10.1086/519795 **[Already cited in Paragraph 14 — second use here]**

Commented [של48]: The LD decay plot (ld_decay_profile_R0.2_line.png) is a main-text figure

Commented [של49]: Quinlan, A. R., & Hall, I. M. (2010). BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics*, 26(6), 841–842. https://doi.org/10.1093/bioinformatics/btq033 **[From My Draft — standard BEDTools citation]**

Commented [של50]: Yates, A. D., Allen, J., Amode, R. M., et al. (2022). Ensembl Genomes 2022: an expanding genome resource for non-vertebrates. *Nucleic Acids Research*, 50(D1), D996–D1003. https://doi.org/10.1093/nar/gkab1007 **[AI Sourced — standard Ensembl Plants citation; verify against your reference manager that this is the correct release-62-era** ...

Commented [של51]: Danecek, P., Bonfield, J. K., Liddle, J., Marshall, J., Ohan, V., Pollard, M. O., Whitwham, A., Keane, T., McCarthy, S. A., Davies, R. M., & Li, H. (2021). Twelve years of SAMtools and BCFtools. *GigaScience*, 10(2), giab008. https://doi.org/10.1093/gigascience/giab008 **[AI Sourced –** ...

Genotype Imputation

To facilitate local linkage disequilibrium estimation for haplotype construction, missing genotypic data were imputed using the sparse Non-negative Matrix Factorization (sNMF) algorithm implemented in the LEA R package v3.6.0 (Frichot & François, 2015). The number of ancestral populations was fixed *a priori* at K = 3, consistent with the three previously described ecotypes of this wild barley collection (Hübner et al., 2009, 2013). The genotypic dataset was first converted from VCF to LFMM format, after which ten independent sNMF runs were performed at K = 3, and the run minimizing the cross-entropy criterion was selected as the optimal model. Missing genotypes were subsequently imputed based on the mode of the ancestral genotype probability distribution derived from the optimal sNMF run, producing a fully homozygous imputed genotype matrix consistent with the predominantly selfing nature of barley. The imputed dataset was then reassembled into VCF format, and gene-level sub-VCFs were extracted from the imputed VCF using the same gene-coordinate intervals (gene start −1 kb to gene end +1 kb) described above, providing the input for linkage disequilibrium estimation within the haplotype analysis pipeline.

Haplotype Construction and Phenotype Association Testing

To test whether haplotype structure within each candidate gene explained phenotypic variation in grain nutritional traits, a haplotype-based association analysis was performed using the crosshap R package 1.4.0 (Marsh et al., 2023). The analysis was applied to all candidate genes identified within the ±220 kb windows of significant GWAS lead SNPs, across all four nutritional traits (protein, starch, beta-glucan, and dietary fiber). For each gene, pairwise linkage disequilibrium was computed as a square pairwise r^2 matrix from the imputed gene-level VCF using PLINK, and local haplotypes were subsequently constructed from the corresponding raw (non-imputed) gene-level VCF using the resulting LD matrix. To probe haplotype resolution at multiple granularities, the haplotyping procedure was run for each gene under two minimum marker-group sizes (MGmin = 2 and MGmin = 3) and across seven clustering threshold (epsilon) values: 0.05, 0.2, 0.4, 0.5, 0.6, 0.8, and 0.85. A minimum haplotype size of nine accessions (minHap = 9) was enforced throughout, and accessions assigned to the unresolved haplotype-zero group were excluded from all downstream statistical tests. The association between haplotype grouping and trait values was evaluated for every gene-by-MGmin-by-epsilon combination using the non-parametric Kruskal-Wallis rank-sum test (Kruskal & Wallis, 1952), constituting the primary discovery criterion.

Multiple Testing Correction and Significance Criteria

Because the parameter sweep across two MGmin values and seven epsilon values generates up to fourteen omnibus tests per gene, a structured two-stage multiple testing correction was applied to control the per-gene and per-trait error rates separately. First, within each gene, tests producing identical results — defined by an identical raw Kruskal-Wallis p-value, number of haplotype groups, and sorted group-size signature — were collapsed into a single representative test, deterministically retaining the smallest epsilon value and, in the event of ties, the larger MGmin. The Holm step-down procedure (Holm, 1979) was then applied across the remaining unique valid tests within each gene, and the test with the smallest within-gene Holm-adjusted p-value was selected as the gene's representative association. Second, separately within each trait, the Benjamini-Hochberg false discovery rate procedure (Benjamini & Hochberg, 1995) was applied to the representative within-gene Holm-

Commented [של52]: **Frichot, E., & François, O. (2015).** LEA: An R package for landscape and ecological association studies. *Methods in Ecology and Evolution*, 6(8), 925–929. https://doi.org/10.1111/2041-210X.12382 **[From My Draft — standard LEA citation]**

Commented [של53]: **•Maybe need to add 2012 or 2013 -> 2012 change needed**

Hübner, S., Höffken, M., Oren, E., Haseneyer, G., Stein, N., Graner, A., Schmid, K., & Fridman, E. (2009). Strong correlation of wild barley (*Hordeum spontaneum*) population structure with temperature and precipitation variation. *Molecular Ecology*, 18(7), 1523–1536. https://doi.org/10.1111/j.1365-294X.2009.04106.x **[From My Draft — already cited in Paragraph 1]**
Hübner, S., Bdolach, E., Ein-Gedi, S., Schmid, K. J., Korol, A., & Fridman, E. (2013). Phenotypic landscapes: phenological patterns in wild and cultivated barley. *Journal of Evolutionary Biology*, 26(1), 163–174. https://doi.org/10.1111/jeb.12043 **[AI Sourced — flagged for Sariel verification; see flag #2 below]**

Commented [של54]: **Marsh, J. I., Johnston, R., Yuan, Y., Cuthbert, J. L., Danilevicz, M. F., Bayer, P. E., Anderson, R., Batley, J., & Edwards, D. (2023).** crosshap: R package for local haplotype visualization for dissection of genetic loci linked to complex traits. *Bioinformatics*, 39(7), btad411. https://doi.org/10.1093/bioinformatics/btad411 **[From My Draft — standard crosshap citation]**

Commented [של55]: ⚠️ **"Across all four nutritional traits" — methodological caveat.** Per your instruction, this matches the Sonnet wording and frames protein as part of the analysis (with the null result handled in Results). However, the actual config.yaml of the All_Genes_All_Traits_V2 pipeline lists only three traits (betaglucan, fiber, starch). A strict reviewer who inspects the GitHub repo could note the discrepancy. **Confirm with Sariel whether the protein analysis from the earlier tool needs a brief Methods mention** (e.g., one sentence in the Results introduction noting that protein was screened but yielded no significant haplotype groupings).

Commented [של56]: **Kruskal, W. H., & Wallis, W. A. (1952).** Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association*, 47(260), 583–621. https://doi.org/10.2307/2280779 **[AI Sourced — standard Kruskal-Wallis citation; verify your bibliography style for older statistical citations]**

Commented [של57]: **Supplementary Material**
Optional supplementary content for this paragraph (not strictly required):
•**Supplementary Table SX (optional).** Per-test statistics table from the script output (per_test_stats.csv), summarizing for each candidate gene the number of unique valid tests retained after duplicate collapsing, the best ...

Commented [של58]: **Holm, S. (1979).** A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. **[AI Sourced — standard Holm 1979 citation]**

Commented [של59]: **Benjamini, Y., & Hochberg, Y. (1995).** Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300. **[Already cited in Paragraph 14; second use here]**

adjusted p-values across all genes; a Bonferroni correction was also computed in parallel for reference. Genes with a trait-level FDR-adjusted p-value below 0.05 were retained as statistically significant, with the FDR-based threshold serving as the primary selection criterion for downstream candidate-gene curation.

Candidate Gene Curation, Wild + Elite Dataset Construction, and Final Figure Generation

The set of statistically significant genes identified in the previous step was further curated through a two-stage filter: functional annotations were obtained from the Ensembl Plants release 62 web portal (Yates et al., 2022) and, where the database annotation was incomplete or putative, were complemented by translated nucleotide-to-protein sequence similarity searches against the NCBI non-redundant protein database using the BLASTX web service (Altschul et al., 1997; Johnson et al., 2008), and only genes with a plausible role in the metabolic pathway underlying the corresponding grain component were retained. This curation yielded a final set of *X* candidate genes spanning three of the four nutritional traits — beta-glucan, fiber, and starch — as no candidate genes for protein passed the statistical significance filter. To contextualize the wild-barley haplotype structure of these candidate genes within a cultivated germplasm reference, six elite cultivated barley lines (Corvette, Triumph, Iris, Gunnar, Manchuria_A, and Rapid) were retrieved from the IPK Barley Pangenome Browser v2 (Jayakodi et al., 2020), a multi-accession barley genomic resource also mapped to the Morex V3 reference genome, on the basis of low missing-data rates across the gene-level variant positions of the *X* candidate genes. For each of the *X* genes, gene-level VCFs spanning the same gene-coordinate intervals used throughout the haplotype pipeline (gene start −1 kb to gene end +1 kb) were downloaded for the six elite lines and merged with the corresponding wild-barley VCF using bcftools v1.13 (Danecek et al., 2021), under a strict-intersection scheme that retained only the SNP positions present in the wild VCF (with elite genotypes absent at these positions encoded as homozygous reference). Final per-gene figures were then generated by re-running the crosshap haplotyping procedure on each merged wild-plus-elite VCF, with linkage disequilibrium estimated from the corresponding imputed wild-barley VCF as described above; for each gene, a single epsilon and MGmin combination was selected to maximize the biological interpretability of the resulting haplotype grouping among the parameter combinations that had yielded statistically significant Kruskal-Wallis associations in the wild-only discovery analysis. The resulting haplotype assignments were used to render an integrated violin-and-heatmap figure assembled in R using the ggplot2 v4.0.1 (Wickham, 2016), ggpubr v0.6.3 (Kassambara, 2023), and patchwork v1.3.2 (Pedersen, 2024) packages. The violin plot displays the per-haplotype-group distribution of the trait BLUP value computed on the 290 wild barley accessions only (with the unresolved haplotype-zero group excluded), annotated with the nominal omnibus Kruskal-Wallis p-value computed on the wild-only haplotype groupings displayed and with significance bars from post-hoc Wilcoxon rank-sum reference-versus-others pairwise comparisons (Holm-corrected), where the reference haplotype group was defined per gene as the largest haplotype group. Below the violin, a genotype heatmap displays, for each haplotype group, a representative wild accession alongside any individual elite cultivar that segregates into that same group, providing a direct visual contrast between the wild allelic structure and the cultivated reference at each candidate locus.

Commented [של60]: **Yates, A. D., Allen, J., Amode, R. M., et al. (2022).** Ensembl Genomes 2022: an expanding genome resource for non-vertebrates. *Nucleic Acids Research*, 50(D1), D996–D1003. **[Already cited in Paragraph 16; second use here]**

Commented [של61]: **Altschul, S. F., Madden, T. L., Schäffer, A. A., Zhang, J., Zhang, Z., Miller, W., & Lipman, D. J. (1997).** Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Research*, 25(17), 3389–3402. **[AI Sourced — foundational BLAST citation]**

Commented [של62R61]: **Johnson, M., Zaretskaya, I., Raytselis, Y., Merezhuk, Y., McGinnis, S., & Madden, T. L. (2008).** NCBI BLAST: a better web interface. *Nucleic Acids Research*, 36(suppl_2), W5–W9. **[From My Draft]**

Commented [של63]: Per your instruction, this matches the Sonnet wording and frames protein as part of the analysis (with the null result handled in Results). However, the actual config.yaml of the All_Genes_All_Traits_V2 pipeline lists only three traits (betaglucan, fiber, starch). A strict reviewer who inspects the GitHub repo could note the discrepancy. **Confirm with Sariel whether the protein analysis from the earlier tool needs a brief Methods mention** (e.g., one sentence in the Results introduction noting that protein was screened but yielded no significant haplotype groupings).

Commented [של64]: **Jayakodi, M., Padmarasu, S., Haberer, G., et al. (2020).** The barley pan-genome reveals the hidden legacy of mutation breeding. *Nature*, 588(7837), 284–289. **[From My Draft]**

Commented [של65]: Satisfying explanation?

Commented [של66]: **Danecek, P., Bonfield, J. K., Liddle, J., et al. (2021).** Twelve years of SAMtools and BCFtools. *GigaScience*, 10(2), giab008. **[Already cited in Paragraph 16; second use here]**

Commented [של67]: **Wickham, H. (2016).** *ggplot2: Elegant Graphics for Data Analysis.* Springer-Verlag, New York. **[From My Draft]**

Commented [של68]: **Kassambara, A. (2023).** ggpubr: 'ggplot2' Based Publication Ready Plots. R package. **[From My Draft]**

Commented [של69]: **Pedersen, T. L. (2024).** patchwork: The Composer of Plots. R package. **[From My Draft]**